

ENGS69 – Ethical and Legal Issues

- ❑ Hacker Ethics/Morality
- ❑ Law and Law Enforcement
- ❑ Privacy vs. Security

Definitions

- ❑ **Ethics:** “The science of morals; the department of study concerned with the principles of human duty. The moral principles by which a person is guided.” – Oxford English Dictionary
- ❑ **Moral:** “Of or pertaining to character or disposition, considered as good or bad, virtuous or vicious; of or pertaining to the distinction between right and wrong, or good and evil, in relation to the actions, volitions, or character of responsible beings; ethical.” – Oxford English Dictionary
- ❑ Terms will be used interchangeably – basically, knowing the difference between right and wrong.

What is a hacker?

- ❑ **Hacker:** “A person with an enthusiasm for programming or using computers as an end in itself.” Or, “A person who uses his skill with computers to try to gain unauthorized access to computer files or networks.” – Oxford English Dictionary
- ❑ The term is misleading and has been used to mean many different things by computer professionals, cyber criminals and the media.
 - Self-described hackers – enjoy experimenting with technology and writing code.
 - Media-labeled hackers (crackers) – break into systems, cause damage, and write malware.
 - Ethical hackers – former hackers or crackers who have joined the security industry to test network security and create security products and services.

Goodies or Baddies?

- ❑ Black Hats – break into systems, develop and share vulnerabilities, exploits, malicious code, and attack tools.
- ❑ Grey Hats – are in hacker 'no-man's land', may work as security professionals by day and 'hack' by night.
- ❑ White Hats – are part of the 'security community', help find security flaws, but share them with vendors so that products can be made safer.
- ❑ There is a fine line between the 'hats' and the distinction often becomes blurred. Often a matter of perspective.



Hacker Ethics

- ❑ Early hackers developed a code of ethics, which has been adopted in large part by computer professionals today.
- ❑ Code of ethics has evolved based on technological and societal changes.
- ❑ Some hackers reject this code for a variety of reasons.



ACM Code of Ethics and Professional Conduct

- 1.1 Contribute to society and human well-being
- 1.2 Avoid harm to others
- 1.3 Be honest and trustworthy
- 1.4 Be fair and take action not to discriminate
- 1.5 Honor property rights including copyrights and patents
- 1.6 Give proper credit for intellectual property
- 1.7 Respect the privacy of others
- 1.8 Honor confidentiality



ACM Code of Ethics and Professional Conduct

2.3 Know and respect existing laws pertaining to professional work.

2.7 Improve public understanding of computing and its consequences.

2.8 Access computing and communication resources only when authorized to do so.

All these rules highlight a need to obey the law, avoid harm, and respect others' privacy and property, but also to further knowledge and understanding.



Ten Commandments of Computer Ethics (CEI)

1. Thou shalt not use a computer to harm other people.
2. Thou shalt not interfere with other people's computer work.
3. Thou shalt not snoop around in other people's computer files.
4. Thou shalt not use a computer to steal.
5. Thou shalt not use a computer to bear false witness.



Ten Commandments of Computer Ethics (CEI)

6. Thou shalt not copy or use proprietary software for which you have not paid.
7. Thou shalt not use other people's computer resources without authorization or proper compensation.
8. Thou shalt not appropriate other people's intellectual output.
9. Thou shalt think about the social consequences of the program you are writing or the system you are designing.
10. Thou shalt always use a computer in ways that insure consideration and respect for your fellow humans.



Alternative Hacker Motivations

- ❑ Extremely diverse community – no single motivation.
 - Freedom of information – progress, advancement of knowledge and science.
 - The Internet as a new political space – lack of legitimate government authority.
 - Fixing security flaws – using one's knowledge to make the Internet safer. Research.
 - No harm, no foul.
 - Notoriety, fame – attaining bragging rights without thinking about consequences.
 - Wish for harm, disruption
 - Profit motive – cybercrime.
 - The Internet as a political tool.



The Hacker's Manifesto

- ☐ Fighting boredom - curiosity
- ☐ Social protest
- ☐ Rebelling against 'mainstream society'
- ☐ Outsmarting the adults - POWER
- ☐ Thirst for knowledge, acceptance, community, excitement, danger, identity

"This is it... this is where I belong.. I know everyone here... even if I've never met them, never talked to them, may never hear from them again... I know you all... we've been spoon-fed baby food at school when we hungered for steak... the bits of meat that you did let slip through were pre-chewed and tasteless... This is our world now... the world of the electron and the switch, the beauty of the baud. We make use of a service already existing without paying for what could be dirt-cheap if it wasn't run by profiteering gluttons, and you call us criminals. We explore... and you call us criminals. We seek after knowledge... and you call us criminals. We exist without skin color, without nationality, without religious bias... and you call us criminals... My crime is that of outsmarting you, something that you will never forgive me for. I am a hacker, and this is my manifesto." – 'The Conscience of a Hacker', The Mentor, Phrack, Volume One, Issue 7, Phile 3, January 8, 1986.



Freedom of Information

- ☐ The spread of information and knowledge should be unrestricted.
- ☐ Progress in human development, philosophy, the arts, and sciences can only occur in a free environment.
- ☐ Leveling the playing field, no more discrimination, being judged on ability, not race, age, religion, background or looks.
- ☐ Closed systems are sinister – they restrict the flow of information - censorship.



The Internet as a New Political Space



- ☐ New political area unencumbered by physical boundaries.
- ☐ Space untouched by the laws of nation-states.
- ☐ New community makes its own rules not governed by existing political or economic structures.
- ☐ Check on government abuse.

No Harm, No Foul

- ❑ It's okay to look around, as long as you don't break anything.
- ❑ Curiosity will help gain a better understanding of how systems work.
- ❑ Is this the only (best) way to learn?



Fixing Security Flaws

- ☐ By breaking into systems one discovers vulnerabilities – these can be fixed, making the systems more secure.
- ☐ Forces unresponsive vendors and companies to act.
- ☐ Community service that protects the Internet against rogues.
- ☐ If the systems aren't properly secured, it's not my fault.
- ☐ Power hunger, arrogance – I know how to do this and you don't.



Notoriety, Fame

- ☐ Hacking for bragging rights – script kiddies.
- ☐ Aren't concerned with damage, or are unaware of the potential consequences of their actions.
- ☐ Hacking is cool!



Causing Harm, Disruption

- ☐ Wish to cause damage, harm
- ☐ No regard for law, morality
- ☐ Anti-social attitudes
- ☐ Loners – grudge against society

Profit Motive

- ❑ No concern for morality
- ❑ Online theft, fraud, identity theft
- ❑ Extortion
- ❑ Money laundering
- ❑ Economic espionage
- ❑ Organized crime



The Internet as a Political Tool

- ❑ Forum for political expression – the ends justify the means
- ❑ Exchange of political ideas – cyber activism
- ❑ Hacktivism
- ❑ Cyberterrorism
- ❑ Information warfare



Morality of Online Activities

- ❑ Online actions can be used to highlight some the issues discussed above.
- ❑ They show that the moral universe on the Internet is blurry and constantly changing.
- ❑ They also show that sometimes morality and law clash.
- ❑ It's not always immediately clear what's right and wrong.



File-Sharing

- ☐ Millions of people worldwide use peer-to-peer (P2P) file-sharing networks (Napster, Gnutella, now Kazaa, Morpheus).
- ☐ Used to acquire and share music, images, video files, and software free.
- ☐ It creates a sense of community.
- ☐ 'Robin Hood' effect – stealing from rich multinational companies.
- ☐ I wouldn't have bought it anyway!
- ☐ The owner doesn't lose possession of his property – reproducibility.
- ☐ What harm does it do? Illegal, but not immoral?
- ☐ Provides exposure for little-known artists.



File-sharing

- ☐ Media and software firms lose billions of dollars per year to file-sharing.
- ☐ Sales of CDs have dropped by 5-10% each of the last two years (RIAA) - because of P2P?
- ☐ It breaks the law, violates copyrights.
- ☐ It endangers R&D into new software, and the production of quality music, movies etc.
- ☐ It raises the price of products for those that legally purchase.
- ☐ It saturates bandwidth.



File-sharing

- ☐ Here's the problem: too many people break the existing laws – does this mean they aren't supported by society and need to be changed?
- ☐ Most people know it's wrong, but it's easy, anonymous, and you probably won't get caught!
- ☐ On the other hand, do companies charge too much for their products? Why does a CD that costs \$1-2 to produce and market cost \$15-20 to buy? Would you pay for it for \$5?
- ☐ Useful and legitimate technologies are being banned because they are being used for illegal purposes. Is this right?
- ☐ What about music companies etc. being allowed to hack you if you use P2P? Do two wrongs make a right?



Full Disclosure

- ❑ Publishing details of software vulnerabilities and exploits – sometimes before vendors are given a chance to fix them.
- ❑ Pros:
 - Helps researchers understand technologies and flaws, furthers knowledge.
 - Helps make products more secure.
- ❑ Cons:
 - Information can be used by hackers to attack systems or write malware.
 - Makes the Internet less secure.
- ❑ Full Disclosure guidelines:
 - 30-day period to allow vendors to develop patches.
 - Need for vendor responsiveness.



Writing Code

- ☐ Thousands of software vulnerabilities discovered every year.
- ☐ Puts users at risk.
- ☐ Is the software vendor responsible for the safety of its code?
- ☐ Minimum secure programming standards.
- ☐ Does software have to be developed with security as a priority?
- ☐ Security vs. functionality.
- ☐ Legal liability for computer software security – like other industries?

0120 245A1 365	J> I'X'hCEQ FONT0
2478 9655 0014	98 I \$&'hCEQ
4789 9987 6524	FONT128 éí
9874 2548 3604	\$&'hCEQ FONT148
7853 1456 0322	μ 'ThCEQ FONT168
0125 3698 1570	<Ü 'ThCEQ FONT071
1248 3965 0147	J> I'X'hCEQ FONT0
3690 5201 7851	I \$&'hCEQ FONT128
0120 245A1 365	éí \$&'hCEQ FONT148
2478 9655 0014	μ 'ThCEQ FONT168
4789 9987 6524	<Ü 'ThCEQ FONT071
9874 2548 3604	J> I'X'hCEQ FONT098
7853 1456 0322	I \$&'hCEQ FONT128
0125 3698 1570	éí \$&'hCEQ FONT148
1248 3965 0147	μ 'ThCEQ FONT168

Unauthorized Intrusions

- ☐ Accessing information contained on a computer without authorization - can be sensitive or even proprietary information.
- ☐ Accessing a system is okay as long as no harm is done. If you leave the door open, I can just walk in!
- ☐ Violation of privacy! Analogy of breaking into a house.
- ☐ Can lead to accidental or intentional loss or destruction of private or critical information (medical records, credit card numbers etc.).
- ☐ Assertion of arrogance – I'll show you how skilled I am and how little you know.
- ☐ Should society distinguish between unauthorized intrusions out of curiosity and for monetary or political gain? How would this work practically and legally?



Web Defacements and Semantic Attacks



- ❑ Website defacements - changing the content of a website for fun or to spread ideals or propaganda.
- ❑ Semantic attacks – change contents in subtle but significant ways.
- ❑ Happens thousands of times each year.
- ❑ Just a bit of fun? What harm does it do?
- ❑ A valid expression of political views?
- ❑ Unauthorized access to someone's system.
- ❑ Costs money in cleanup costs, lost productivity and loss of reputation.
- ❑ Could be used to spread false information – financial gain or to cause panic.

Distributed Denial of Service (DDoS) Attacks

- ❑ Use of armies of 'zombie' machines to send enormous amounts of data packets to flood the resources of a website or other service.
- ❑ Renders the computer inaccessible to legitimate traffic.
- ❑ Examples of DDoS Attacks:
 - CNN, Yahoo!, eBay, Amazon etc., February 2000 – 'Mafiaboy'.
 - 13 Root Domain Name Servers (DNS), October 2002.



DDOS Attacks

- ❑ Started as pranks to get one's name in the headlines.
- ❑ Again, what's the problem, no physical damage was caused?
- ❑ Reports of massive 'zombie' armies.
- ❑ Serious real-world effects:
 - Loss of business or reputation.
 - Security costs.
 - Disruption of communications.
 - Potential to disrupt crucial national security communications and military operations. (On Sept. 11, the Internet was the only way for some first responders to communicate.)



Worms and Viruses



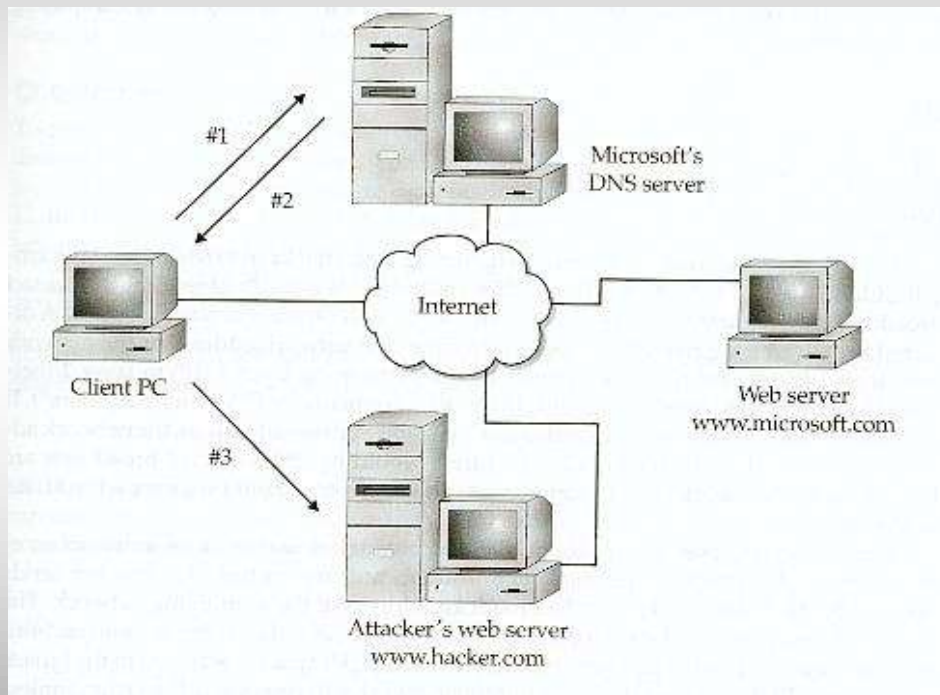
- ☐ Malicious code (malware), replicating or self-replicating computer programs.
- ☐ Multiple potential infection vectors (software flaws, social engineering), propagation methods (e-mail, file-sharing), and payloads (delete or extract data, install other malware).
- ☐ Have the potential to infect millions of machines worldwide in a few minutes.

Worms and Viruses

- ☐ Script kiddies showing off.
- ☐ Most don't have a dangerous payload.
- ☐ Could be a valid form of political expression?
- ☐ Could help identify security flaws – worms that fix vulnerabilities!
- ☐ Billions of dollars in damages and clean-up costs (Melissa, Code Red, MyDoom).
- ☐ Cause disruptions even if the payload is harmless (Slammer, Blaster).
- ☐ Trend towards viruses written for financial gain (spreading spam).
- ☐ Potential for truly malicious payloads that could have national security implications.



Internet Infrastructure Attacks



❑ Attacks against the domain name system (DNS) or Internet routers.

- No valid justification.
- Attacks could slow down, misdirect or halt Internet traffic.
- Could cost billions of dollars.
- Possible national security implications.

Critical Infrastructures- Compound Attacks

- ❑ Critical infrastructures include gas, power, water, banking and finance, transportation, communications.
- ❑ All dependent to some degree on information systems, and automated control systems (SCADA).
- ❑ Employ some or all of aforementioned cyber attacks techniques.
- ❑ Possibly combined with conventional (physical) terror attacks.
- ❑ Consequences include devastating disruption in communication and commerce, blocked access to critical systems, or loss of life.



Online Activism, Hacktivism or Cyberterrorism?

- ☐ The Internet has become a powerful tool for political expression.
- ☐ There is a spectrum of politically-motivated activity online – goes from the legitimate to the potentially disastrous.
- ☐ The Internet is being used to exchange views, plan and coordinate action, and disrupt events and communications.
- ☐ In future, it may be used to cause widespread damage and disruption, loss of life, and political upheaval.
- ☐ Almost all major conflicts also have a cyber component.



Online Activism

- ☐ Politically-minded people collect information online.
- ☐ They exchange information via e-mail, chat rooms, instant messaging etc.
- ☐ They petition politicians and other decisionmakers online.
- ☐ They express their views online.
- ☐ They organize and coordinate protests using the Internet.
- ☐ This is all legal and moral. In fact, in a democracy, it's desirable.



Hacktivism

- ☐ Active electronic protests on the Internet.
- ☐ Denial of service.
 - Electronic sit-ins.
 - E-mail bombs.
- ☐ Unauthorized intrusions – hacking, website defacements.
- ☐ Is this still an acceptable form of political expression? Civil disobedience?
- ☐ The boundary of legality is crossed, but what about morality? Freedom of speech?



Cyberterrorism

- ❑ Still a theoretical construct.
- ❑ Terrorist groups, including al Qaeda, are using the Internet to communicate, recruit, spread propaganda, and plan attacks.
- ❑ Terrorists may be developing cyberwarfare capabilities. Terrorist sympathizers.
- ❑ Foreign nations are developing cyber attack capabilities (i.e. information warfare).
- ❑ What are the implications of this? Are these valid forms of political expression? Do nation-states have the right to use cyber attacks as military tools? What are the implications? How do cyber attacks compare morally with other military means?



Hacking is unethical

- ☐ Causes harm.
- ☐ Denies access to resources or services.
- ☐ Violates property and privacy rights.
- ☐ Allows for no control or accountability.
- ☐ Would they want you to do this to them?
- ☐ Difference between criminal hackers and script kiddies – intent to profit.



Changing Attitudes

- ☐ Leading by example. Living by an ethical code.
- ☐ Empowering people. Education. Responsibility and inclusion.
- ☐ Making people understand the consequences of their actions. Taking responsibility.
- ☐ Changing practices and social norms over time. Netiquette.
- ☐ Legislation. Law enforcement.

Cyberlaw - Difficulties

- ❑ Internet still relatively 'lawless'.
- ❑ Cyberlaw is constantly changing, evolving, being shaped through experience.
- ❑ Internet doesn't fit into traditional notions of territoriality and state boundaries – internationalization of cybercrime.
- ❑ The Internet creates jurisdictional problems.
- ❑ Uniform laws don't exist everywhere.



'Lawless Internet?'

- ☐ The Internet is not nearly as heavily regulated as the physical world.
- ☐ Cybercrimes and hacking were virtually ignored by the law until the mid-1980's.
- ☐ There has subsequently been legislation, such as the Computer Fraud and Abuse Act, the Copyright Act, the Electronic Communications Privacy Act, the USA Patriot Act, and the CAN SPAM Act.
- ☐ Compared to the physical world, many areas remain unregulated or ambiguous.



Many Jurisdictions

- ❑ Who's responsible for policing the Internet? States, nation-states or international entities? Need for a separate jurisdiction for the Internet?
- ❑ There are no governing bodies on the Internet – just standards entities (ICANN, NANOG etc.).
- ❑ International cooperation in cybercrime and cyber attack investigations (real-time?).
- ❑ Still old 'territorial' problems – Where is the hacker? Where is the server based? Through how many countries does the data pass? Which law do you use?



No Unifying Laws

- ❑ Different countries (about 60) have different cyber laws – some don't have any (ILoveYou).
- ❑ Laws can be conflicting or contradictory (extradition requires dual criminality).
- ❑ Laws can be weak or poorly enforced.
- ❑ Attempts to create international legislation – (European) Convention on Cybercrime.



Law Enforcement

- ❑ Insufficient reporting (CSI/FBI report – only 30% of companies that experienced computer intrusions reported them to law enforcement).
- ❑ Inadequate cooperation between the private sector and law enforcement (FOIA), and between different countries.
- ❑ Lacking cyber investigation and law enforcement skills and knowledge.
- ❑ Lacking tools and technologies.



Criminal Justice Changes – Areas of Critical Need

❑ “Electronic Crime Needs Assessment for State and Local Law Enforcement” – DOJ/NIJ.

- 1: Public Awareness
- 2: Data and Reporting
- 3: Uniform Training & Certification Courses
- 4: Onsite Management Assistance for Electronic Crime Units and Task Forces
- 5: Updated Laws
- 6: Cooperation with High-tech Industry
- 7: Special Research and Publications
- 8: Management Awareness and Support
- 9: Investigative and Forensic Tools
- 10: Structuring a Computer Crime Unit



Criminal Justice Changes – Areas of Critical Need

- ❑ “Law Enforcement Tools and Technologies For Investigating Cyber Attacks” – ISTS
 - Preliminary Investigation and Data Collection (data collection, recovery, mapping, analysis and storage tools).
 - Log Analysis (transferable logs across formats and platforms).
 - IP Tracing and Real-Time Intercepting (catching attacks as they happen).
 - Research and Development into Emerging Technologies (tools to circumvent encryption, recognize unauthorized wireless network access and steganography).
 - National Information Sharing (data, contacts and expertise).
 - Law Enforcement-Specific Issues (tools for law enforcement needs; ability to match products with capabilities).
 - Training (law enforcement programs and benchmarks).



Law Enforcement Successes

- ❑ Robert Tappan Morris Jr. – the Morris Worm Nov. 1988 – convicted under the 1986 Computer Fraud and Abuse Act to 3 years probation, 400 hours community service, and \$10,000 fine.
- ❑ Kevin Mitnick – hacking spree – pled guilty in 1999 to various counts of wire fraud, computer fraud and illegally intercepting a wire communication; sentenced to 46 months in prison, \$4,125 restitution, 3-year supervised release.
- ❑ “Mafiaboy” – DoS attacks in 2000 – 15-year old Canadian arrested in April 2000; charged with 12 counts of ‘mischief to data’ and 54 counts of illegal use of computer systems; pleaded guilty to 56 of the 66 charges; sentenced to 8 months youth detention and 1 year of probation.



Law Enforcement Successes

- ❑ David Smith – the Melissa Virus March 1999 – arrested in April 1999; in May 2002, sentenced to 20 months in jail and a \$5,000.
- ❑ Jan de Wit - Kournikova Virus Feb. 2001 – sentenced to 150 hours of community service in the Netherlands in September 2001.
- ❑ Gary McKinnon - British hacker broke into government and DoD systems in 2001 – charged in Nov. 2002 with multiple counts of unauthorized access and damage (\$900,000) to federal computer systems; faces up to 5 years in prison and a \$250,000 fine.
- ❑ Daniel J. Baas – Hacked into Axcion Corp. Systems and stole customer information between Jan. 2001 and Jan. 2003 – charged in August 2003; could face up to 5 years in prison and a \$250,000 fine.
- ❑ Adrian Lamo – Hacked into the computer network of the New York Times in 2002 – pled guilty in Jan. 2004 to one count of computer damage; faces up to 1 year in prison.



Law Enforcement Successes – A Drop in the Ocean?

- ❑ Laws and law enforcement have made headway in recent years worldwide in finding, prosecuting and convicting hackers and cybercriminals.
- ❑ The deck is still stacked in hackers' favor – Internet offers anonymity; physical distance; lacking, ambiguous or competing laws and jurisdictions; and plenty of vulnerabilities and easily-available attack tools.
- ❑ Law enforcement seldom acts against cyber offenses unless a certain amount of 'monetary damage' (\$5,000 in the U.S. for federal charges) has occurred – even then, enforcement is patchy.
- ❑ Punishments are relatively minor and not a good deterrent – this is starting to change.
- ❑ Major (costly cyber attacks) are still unresolved: Code Red, Nimda, Slammer, Blaster, Sobig, MyDoom, DDoS against the DNS root servers.



Cyber Law

- ☐ Access Device Fraud
- ☐ 1986 Computer Fraud and Abuse Act
- ☐ Wire Fraud
- ☐ Mail Fraud
- ☐ Identity Theft and Assumption Deterrence Act
- ☐ Criminal Infringement of Copyright
- ☐ Obscenity and Child Pornography Laws



Cyber Law – The USA

PATRIOT Act

- ❑ The 2001 USA PATRIOT (Provide Appropriate Tools Required to Intercept and Obstruct Terrorism) Act allows law enforcement to better monitor communications, including e-mail and cell phone; allows for 'roving wiretaps'.
- ❑ Expands existing surveillance powers to apply to electronic communications.
- ❑ Allows ISPs to report customer information when they believe there is "immediate danger of death or serious physical injury to any person."
- ❑ Law enforcement can monitor hacker (unauthorized user) activities without a warrant if the information is believed to be relevant to an investigation.
- ❑ Pen register (message headers) and wiretap (message contents) warrants are still required to monitor legitimate ISP customers.



Cyber Law – The USA

PATRIOT Act

- ☐ Patriot Act is not specific to terrorist hackers, but can be applied to any computer crime investigation.
- ☐ Allows for information sharing between law enforcement and intelligence agencies.
- ☐ Expands definitions of computer crimes and online terrorism – imposes harsher sentences. Cyberterrorism that endangers life or public health can be punishable by life imprisonment.
- ☐ Patriot Act has been criticized for being rushed, draconian and violating privacy rights.
- ☐ 'Sunset' provisions and reporting mechanisms exist, but oversight is secretive.
- ☐ Attempts to turn back the clock, change provisions of the Act.
- ☐ Plans for a Patriot Act II, heavily opposed.



Cyber Law – The DMCA

- ❑ The 1998 Digital Millennium Copyright Act (DMCA) provides copyright holders protection with regards to digital communications systems.
 - “In general terms, copyright provides an author with a tool to protect a work from being taken, used, and exploited by others without permission.”
 - Copyrights allow the holder to control access to, copying of, and distribution of his protected material, except concerning ‘fair use’.
- ❑ DMCA clarifies copyright law on the Internet.
- ❑ Copyrights for original works are protected – in contrast to the physical world, the Internet allows for a ‘product’ to be copied and distributed widely, while allowing the owner of the ‘product’ to retain possession.
- ❑ Fighting the pirates. File-sharing and copying software are explicitly outlawed, as are technologies that circumvent copyright protection mechanisms.



Cyber Law – The DMCA

- ❑ DMCA is highly contested for several reasons (Ed Felton, Dmitry Sklyarov):
 - Stifles free expression and scientific research.
 - Jeopardizes 'fair use'.
 - Impedes competition and innovation.
- ❑ DMCA is at the center of the music industry's legal battle against file-sharers.
- ❑ Critics want the DMCA amended.

(European) Convention on Cybercrime

- ❑ International treaty to codify cybercrime laws and regulations internationally.
- ❑ As of Jan. 30, 2004, 33 members of the Council of Europe (including France, Germany, Italy and the UK), as well as the US, Canada, Japan and S. Africa have signed the treaty.
- ❑ It will take force as soon as it is ratified by 5 members (at least 3 from the Council of Europe).
- ❑ So far, ratified by Albania, Croatia, Estonia and Hungary.



(European) Convention on Cybercrime

- ❑ Negotiated by members of the Council of Europe, the US, Canada, Japan and S. Africa to create “a common criminal policy aimed at the protection of society against cybercrime.”
- ❑ Established regulations concerning crimes committed via the Internet and other computer networks. Deals with:
 - Copyright infringements
 - Computer-related fraud
 - Child pornography
 - Computer system and network attacks
 - An additional protocol will deal with the publication of racist and xenophobic material via computer networks.
- ❑ Establishes law enforcement and legal powers and procedures for the search of computers and the interception of data traffic.



The CAN-SPAM Act

- ❑ The CAN-SPAM (Controlling the Assault of Non-Solicited Pornography and Marketing) Act, a federal anti-spam law, went into effect January 1, 2004.
 - Prohibits the use of false header information in bulk commercial e-mail.
 - Requires unsolicited messages to contain valid contact and opt-out information.
 - Prohibits harvesting e-mail addresses from websites.
 - Penalties for violations include fines of up to \$250 per e-mail, capped at \$6 million.
- ❑ Critics fear it won't curb the flow of spam because most spammers are based outside the US (similar laws are missing in most countries), enforcement of the law will be difficult, and the Act overrides stricter state laws (California).



Privacy

- ❑ Privacy rights are not explicitly secured in the Constitution or the Bill of Rights.
- ❑ They are usually derived from the Fourth Amendment – the right against unreasonable searches and seizures.
- ❑ Privacy is the power to control what other people know about you:
 - Information about you that has been revealed to the public (poorly protected).
 - Information about you that has been kept private (quite well protected).



Privacy

- Societal and technological advances have changed the nature of privacy in three ways:
 - Scale of information gathered.
 - Kind of information gathered.
 - Scale of exchange of information gathered.

“We can collect, store, manipulate, exchange, and retain practically infinite quantities of data.” – ‘Computer Ethics’, Deborah G. Johnson, 2001

Privacy – The Growing Threat?

□ Is our society turning into 'Big Brother'?

- Huge quantities of personal information are stored in multiple government and corporate databases.
- Abuses of private data are not unheard of.
- Many examples of unauthorized intrusions (hacking) into government and private databases containing names, addresses, credit card numbers, SSNs, medical information, etc.
- Many examples of hacking into PCs.
- Trojans, keystroke loggers, spyware etc.



“As automation increasingly invades modern life, the potential for Orwellian mischief grows.” – Supreme Court Justice Ginsberg.



Privacy – The Growing Threat?

- ❑ What information is out there about you?
 - Financial data: bank transactions, credit history, mortgage, salary etc.
 - Interaction with government(s): SSN, driver's license, taxes, visa applications, criminal record.
 - Medical information: medical history, doctor's visits, medication, operations, health issues.
 - Communications data: landline and cell phone usage (including location!), e-mail messages, websites visited, online shopping.
 - Financial transactions: credit card transactions, purchases (often with details), deposits and withdrawals etc.
 - Travel details: places visited, means of transportation, routes etc.
 - Miscellaneous: All sorts of other info, including reading habits, hobbies etc.



Privacy – The Growing Threat?

- ❑ What could all this information reveal?
 - Your whereabouts over different periods of time (hours, days, months, years).
 - Your financial situation.
 - Your personal life.
 - Your daily habits and routine.
 - Your preferences, likes and dislikes.
- ❑ If all the data were centrally collected it would be SCARY!!!

Securing Privacy

- ❑ Much has been done to protect privacy
 - legislation, advocacy and practical measures.
- ❑ Privacy legislation:
 - Fair Credit Reporting Act of 1970
 - Privacy Act of 1974
 - Privacy Protection Act of 1980
 - Electronic Communications Privacy Act of 1986
 - Right to Financial Privacy Act
 - Federal Records Act
 - Health Information Portability and Accountability Act (HIPAA) of 1996
 - Gramm-Leach-Bliley Act of 1999
 - Myriad state privacy laws (California's SB 1386 Privacy Law)



Securing Privacy

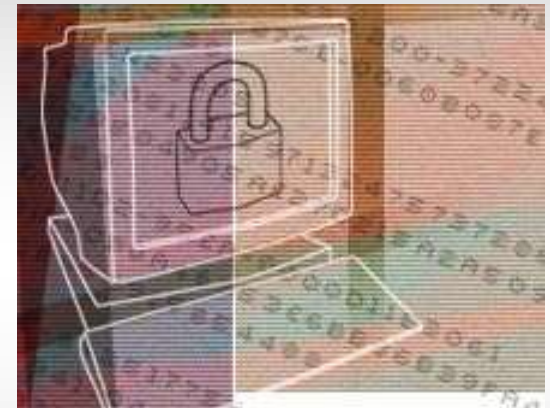
❑ Key Privacy Issues (reflected in most privacy laws):

- Databases or data collection should not exist in secret.
- Individuals must be able to find out what information is being stored about them and how it is being used.
- Individuals should be able to prevent information stored for one purpose being used for another.
- Individuals should be able to correct inaccurate information stored about them.
- Organizations collecting information must make efforts to check the reliability of their information and prevent its misuse.



Privacy vs. Security

- ❑ There has always been a delicate balance between privacy and security.
- ❑ The balance has to be re-negotiated whenever political, societal or technological factors change.
- ❑ Sept. 11, 2001 and the ongoing war on terrorism have forced a re-evaluation of the privacy vs. security balance.



The Case for Security

- ❑ Terrorists have the upper hand on the Internet – anonymity (re-mailer, encryption, steganography).
- ❑ Terrorists use the Internet (and other communications systems) to recruit, spread propaganda, plan and coordinate attacks, and, perhaps soon, to launch attacks.
- ❑ Criminals (and organized crime groups) use the same security holes to commit fraud, identity theft, and other online offenses.
- ❑ Collecting data has other benefits, such as improving government efficiency and services, or private sector services, and fighting crime.



Privacy vs. Security

- ❑ Government measures to improve security include:
 - Legislation – The USA PATRIOT Act.
 - Increased Surveillance of communications and public places (Carnivore, Echelon, Biometrics).
 - Collection of personal information from multiple government and private databases in search of terrorist patterns. Abandoned DoD Total (Terrorist) Information Awareness Project.
 - US VISIT – Terrorist Watch Lists
- ❑ Are these measures eroding civil liberties and freedom? Congressional and public opposition.



Why Worry About Privacy?

- ☐ Who cares whether this data is stored about me if I haven't done anything wrong?
- ☐ Out of principle people have a right to privacy in a free society.
- ☐ The information could be accessed by unauthorized people/organizations.
- ☐ The information could be altered.
- ☐ Many important decision are made every year based on this information (loans, credit cards, mortgages, employment, housing, health care, law enforcement, national security).
- ☐ Personal privacy fosters trusting relationships – key tenet of democracy.

Finding the Privacy Balance

- ❑ So much data is already collected – can we build in safeguards that let us use it AND protect our freedoms?
- ❑ Legislation – would harmonizing state laws (and international privacy laws) offer additional safeguards?
- ❑ Technology – cryptography can help protect some sensitive online transactions and data, anonymizers can protect one's online identity, and security tools can block access to databases.
- ❑ Responsible government – in times of crisis, people look to government for leadership. Government must act responsibly and not abuse power. Oversight to ensure they don't!



Questions

- ☐ Can you think of a situation where it would be okay to access a computer system without authorization? Discuss.
- ☐ Explain why hacking is not only illegal, but also unethical.
- ☐ Why do people hack? Discuss different motivations for breaking into systems or writing malicious code.
- ☐ Is file-sharing only morally wrong because it is illegal? Discuss.
- ☐ Are virus writers pranksters or criminals? Discuss.
- ☐ Where is the ethical boundary for political action online? Discuss.

Questions

- ☐ Why is it difficult to catch or prosecute hackers? Discuss.
- ☐ What are the benefits and drawbacks of the Patriot Act (or the DMCA)? Discuss.
- ☐ What is the main challenge to successful law enforcement on the Internet?
- ☐ In what ways do technology and the Internet jeopardize privacy?
- ☐ Why is keeping personal information private important? Discuss.
- ☐ Does the war on terrorism require a loss of privacy?



Resources

□ Books:

- 'Computer Ethics, Third Edition', Deborah G. Johnson, Prentice Hall, 2001
- 'Computer Ethics – Cautionary Tales and Ethical Dilemmas in Computing', Tom Forester and Perry Morrison, The MIT Press, 1990
- 'Crypto Anarchy, Cyberstates, and Pirate Utopias', edited by Peter Ludlow, The MIT Press, 2001
- 'CyberLaw – The Law of the Internet', Jonathan Rosenoer, Springer Verlag, 1997
- 'The Hacker Ethic', Pekka Himanen, Random House, 2001
- 'Privacy on the Line: The Politics of Wiretapping and Encryption', Whitfield Diffie and Susan Eva Landau, MIT Press, 1998
- 'Database Nation: The Death of Privacy in the 21st Century', Simson Garfinkel, O'Reilly and Associates, 2001
- 'The Hacker Ethic', Pekka Himanen, Random House, 2001
- 'Computers, Ethics & Social Values', Edited by Deborah G. Johnson and Helen Nissenbaum, Prentice Hall, 1995
- 'Virtual States – The Internet and the Boundaries of the Nation-State', Jerry Everard, Routledge, 2000
- 'Hacker Culture', Douglas Thomas, University of Minnesota Press, 2002
- 'Cybering Democracy – Public Space and the Internet', Diana Saco, University of Minnesota Press, 2002



Resources

❑ Online Resources:

- The Brookings Institution, Computer Ethics Institute – http://www.brook.edu/its/cei/cei_hp.htm
- Ten Commandments of Computer Ethics - http://www.brook.edu/dybdocroot/its/cei/overview/Ten_Commandments_of_Computer_Ethics.htm
- 'ACM Code of Ethics and Professional Conduct', Association for Computing Machinery, October 16, 1992 - <http://www.acm.org/constitution/code.html>
- Electronic Frontier Foundation - <http://www.eff.org/>
- 'Responsible Vulnerability Disclosure Process – Internet Draft', Steve Christey and Chris Wysopal, 2002 - http://www.whitehats.ca/main/about_us/policies/draft-christey-wysopal-vuln-disclosure-00.txt
- 'Cyber-Safety for Everyone: From Kids to Elders', M.E. Kabay - <http://www2.norwich.edu/mkabay/cyberwatch/cybersafety.pdf>
- 'Cyberspace Law for Non-Lawyers', Larry Lessig, David Post and Eugene Volokh - http://www.eff.org/Government/Legislation/Legal/CyberLaw_Course/
- 'The Legal Framework – Unauthorized Access to Computer Systems, Penal Legislation in 44 Countries', Stein Schjolberg, Last Updates April 7, 2003 - <http://www.mossbyrett.no/no/info/legal.html>
- 'Total "Terrorism" Information Awareness (TIA) – Latest News' – Electronic Privacy Information Center - <http://www.epic.org/privacy/profiling/tia/>
- Cybercrime Law Website - <http://www.cybercrimelaw.net/>
- State Laws Related to Internet Privacy Website, National Conference of State Legislatures - <http://www.ncsl.org/programs/lis/privacy/eprivacylaws.htm>



Resources

□ Reports/Press Releases:

- 'Cyber Attacks During the War on Terrorism – A Predictive Analysis', The Institute for Security Technology Studies at Dartmouth College, September 22, 2001 - http://www.ists.dartmouth.edu/ISTS/counterterrorism/cyber_a1.pdf
- 'Law Enforcement Tools and Technologies for Investigating Cyber Attacks', Institute for Security Technology Studies at Dartmouth College, 2002 - <http://www.ists.dartmouth.edu/TAG/lena.htm>
- '2003 Computer Crime and Security Survey', Computer Security Institute (CSI), May 2003 - <http://www.gocsi.com/press/20030528.jhtml>
- 'Electronic Crime Needs Assessment for State and Local Law Enforcement', National Institute of Justice, U.S. Department of Justice, March 2001 - <http://www.ncjrs.org/pdffiles1/nij/186276.pdf>
- 'CYBERTERRORISM', Testimony before the Special Oversight Panel on Terrorism Committee on Armed Services, U.S. House of Representatives, by [Dorothy E. Denning](#), Georgetown University, May 23, 2000 - <http://www.cs.georgetown.edu/~denning/infosec/cyberterror.html>
- 'Kevin Mitnick Sentenced to Nearly Four Years in Prison; Computer Hacker Ordered to Pay Restitution to Victim Companies Whose Systems Were Compromised', U.S. Department of Justice, United States Attorney's Office Central District of California, Press Release, August 9, 1999 - <http://www.usdoj.gov/criminal/cybercrime/mitnick.htm>

Resources

□ Reports/Press Releases:

- 'Security Vulnerability Reporting and Response Process – Public Review Version', Organization for Internet Safety, June 4, 2003 - <http://www.oisafety.org/process.html>
- 'USAPA Sunset Provisions Could Leave Congress in the Dark - Ongoing Oversight of Anti-Terrorism Law Called For', Electronic Frontier Foundation - http://www.eff.org/Privacy/Surveillance/Terrorism_militias/20011212_eff_usapa_sunset_analysis.html
- 'Son-of-Patriot Act', Electronic Frontier Foundation - <http://www.eff.org/Privacy/Surveillance/Terrorism/son-of-patriot.php>
- 'Unintended Consequences: Five Years Under the DMCA', Electronic Frontier Foundation, 2003 - http://www.eff.org/IP/DMCA/20030102_dmca_unintended_consequences.html
- 'Civil Liberties After 9/11 – The ACLU Defends Freedom', The American Civil Liberties Union, September 20, 2002 - <http://www.aclu.org/SafeandFree/SafeandFree.cfm?ID=10898&c=207>



Resources

□ Papers:

- 'Activism, Hacktivism, and Cyberterrorism: The Internet as a Tool for Influencing Foreign Policy', Dorothy E. Denning, Georgetown University, [Internet and International Systems: Information Technology and American Foreign Policy Decisionmaking Workshop](http://www.nautilus.org/info-policy/workshop/papers/denning.html), 1999 - <http://www.nautilus.org/info-policy/workshop/papers/denning.html>
- 'Cyberwarriors - Activists and Terrorists Turn to Cyberspace', DOROTHY DENNING, Harvard International Review, Vol. XXIII, No. 2, Summer 2001, pp. 70-75. - <http://www.cs.georgetown.edu/~denning/infosec/harvard.final.rtf>
- 'Is Cyber Terror Next?', [Dorothy E. Denning](http://www.ssrc.org/sept11/essays/denning.htm), Professor of Computer Science; Director of the Georgetown Institute for Information Assurance, Georgetown University, Social Science Research Council, November 2001 - <http://www.ssrc.org/sept11/essays/denning.htm>
- "Disarming the Back Hats – When does a security tool become a cyber weapon?", Dorothy Denning - <http://www.cs.georgetown.edu/~denning/infosec/disarming-blackhats.html>
- 'The Critical Challenges from International High-Tech and Computer-Related Crime at the Millenium', Michael A. Sussmann, Duke Journal of Comparative and International Law, Vol. 9:45, Spring 1999 - <http://www.law.duke.edu/shell/cite.pl?9+Duke+J.+Comp.+&+Int'l+L.+451>

Resources

□ Papers:

- 'The Morris Worm: how it Affected Computer Security and Lessons Learned by it', Larry Boettger, SANS Reading Room, December 24, 2000 - http://www.giac.org/practical/gsec/Larry_Boettger_GSEC.pdf
- 'The 2001 Patriot Act and Its Implications for the IT Security Professional', Oscar W. Peterson III, SANS Reading Room, February 16, 2002 - http://www.sans.org/rr/legal/patriot_act2.php
- 'Mission Impossible: Warrants in the New Millennium', Deborah A. Orr, SANS Reading Room, May 20, 2001 - http://www.giac.org/practical/gsec/Deborah_Orr_GSEC.pdf
- 'Full Disclosure', Brian Deline, SANS Reading Room, November 28, 2000 - http://www.giac.org/practical/gsec/Brian_DeLine_GSEC.pdf
- 'Defenders or Digilantes?', Christopher Loomis, SANS Reading Room, December 14, 2000 - http://www.giac.org/practical/gsec/Christopher_Loomis_GSEC.pdf
- 'Ethics of Hacktivism', Julie L.C. Thomas, SANS Reading Room, January 12, 2001 - http://www.giac.org/practical/gsec/Julie_Thomas_GSEC.pdf
- 'A Separate Jurisdiction for Cyberspace?', Juliet M. Oberding and Terje Norderhaug - <http://www.ascusc.org/jcmc/vol2/issue1/juris.html>
- 'Using Security to Protect the Privacy of Customer Information', Alan Pachoca, SANS Reading Room, February 21, 2002 - <http://www.sans.org/rr/papers/index.php?id=690>
- 'A Survey of Recent Threats to Privacy Rights', Richard Gutter, SANS Reading Room, January 23, 2002 - <http://www.sans.org/rr/papers/index.php?id=689>



Resources

□ Papers:

- 'Crossing the Line: Ethics for the Security Professional', Scott Carle, SANS Reading Room, March 21, 2003 - <http://www.sans.org/rr/papers/index.php?id=890>
- 'Preparing for HIPAA: Privacy and Security Issues to be Considered', Sherry Fischer, SANS Reading Room, January 5, 2003 - <http://www.sans.org/rr/papers/index.php?id=899>
- 'The Legal System and Ethics in Information Security', Amit Raju Philip, SANS Reading Room, July 15, 2002 - <http://www.sans.org/rr/papers/index.php?id=54>
- 'An Uneven Playing Field: The Advantages of the Cyber Criminal vs. Law Enforcement – and Some Practical Suggestions', Torri Piper, SANS Reading Room, September 10, 2002 - <http://www.sans.org/rr/papers/index.php?id=115>
- 'Full Disclosure and the Window of Exposure', Counterpane Security - <http://www.schneier.com/crypto-gram-0009.html#1>
- 'Is There a Hacker Ethic for 90s Hackers?', Steven Mizrach - <http://www.fiu.edu/~mizrachs/hackethic.html>
- 'Computer Hacking and Ethics', Brain Harvey, University of California, Berkeley - <http://www.cs.berkeley.edu/~bh/hackers.html>
- 'Totem and Taboo in Cyberspace: Integrating Cyberspace into our Moral Universe – Fourth Edition', M.E. Kabay, April 2001 - <http://www.developer.com/tech/article.php/782681>
- 'When Cyber Hacktivism Meets Cyberterrorism', Larisa Paul, February 19, 2001 - http://www.giac.org/practical/gsec/Larisa_Paul_GSEC.pdf
- 'Taking a stand on hackers', Dr. E. Eugene Schultz, Computers & Security, Vol. 21, Issue 5, October 1, 2002
- 'Would Professor Moriarty Make a Credible Sherlock Holmes?', Berni Dwan, Computers & Security, Vol. 2001, Issue 8, August 1, 2001



Resources

□ News Articles:

- 'Patriot Act' Opens ISPs to Monitoring', George A. Chidi Jr., IDG News Service November 05, 2001 - <http://www.pcworld.com/news/article/0,aid,69662,00.asp>
- 'ACLU Sues for Details on Federal Snooping', Michelle Madigan, PC World, October 24, 2002 - <http://www.pcworld.com/news/article/0,aid,106338,00.asp>
- 'Feds Get New Snooping Powers', Scarlet Pruitt, IDG News Service, November 20, 2002 - <http://www.pcworld.com/news/article/0,aid,107136,00.asp>
- 'ACLU Campaign Challenges Patriot Act', Michelle Madigan, PC World, October 16, 2002 - <http://www.pcworld.com/news/article/0,aid,106002,00.asp>
- 'E-Terrorism: Liberty vs. Security', John Borland and Lisa m. Bowman, ZDNet, August 27, 2002 - <http://zdnet.com.com/2100-1105-955493.html>
- 'In Terror War, Privacy vs. Security', Robert O'Harrow Jr., Washington Post, June 3, 2002 - <http://online.securityfocus.com/news/461>
- 'Yes, You Are Being Watched', Stephen Lawson, IDG News Service, December 27, 2002 - <http://www.pcworld.com/news/article/0,aid,108121,pg.1,00.asp>
- 'Total Information Awareness official responds to criticism', Shane Harris, GovExec.com, January 31, 2003 - <http://www.govexec.com/dailyfed/0103/013103h1.htm>
- 'OIS Releases Vulnerability Reporting Plan', Dennis Fisher, eWeek, June 5, 2003 - <http://www.eweek.com/article2/0,4149,1494595,00.asp>



Resources

□ News articles:

- 'CERT, Feds Consider New Reporting Process', Dennis Fisher, eWeek, March 24, 2003 - <http://www.eweek.com/article2/0,4149,1500295,00.asp>
- 'All Eyes on Total Info Awareness', [Dan Caterinicchia](#), Federal Computer Week, December 16, 2002 - <http://www.fcw.com/fcw/articles/2002/1216/web-tia-12-16-02.asp>
- 'Bills: Down With Citizen Database', Ryan Singel, Wired News, January 17, 2003 - <http://www.wired.com/news/privacy/0,1848,57263,00.html>
- 'Security Law Puts Internet Under Scrutiny', Ted Bridis, Associated Press, November 20, 2002 - <http://www.detnews.com/2002/technology/0211/20/a05-15520.htm>
- 'House cyberterrorism bill would protect ISPs', Cara Garretson, Infoworld, February 13, 2003 - <http://archive.infoworld.com/articles/hn/xml/02/02/13/020213hncyberbill.xml>
- 'Antipiracy bill finally sees Senate', John Borland, C-net, March 21, 2002 - <http://news.com.com/2100-1023-866337.html>
- 'Music Biz Lament: Stealing Hurts', Brad King, Wired News, September 26, 2002 - <http://www.wired.com/news/mp3/0,1285,55393,00.html>
- 'Strikeback, Part Deux', Tim Mullen, SecurityFocus, January 13, 2003 - <http://online.securityfocus.com/columnists/134>
- 'Rep. Goodlatte Calls For "War" Against Digital Piracy', Washington Post, April 18, 2002 - <http://www.washingtonpost.com/ac2/wp-dyn?pagename=article&node=&contentId=A10857-2002Apr18¬Found=true>
- 'Microsoft to hackers: don't publish code', Robert Lemos, C-net, October 17, 2001 - <http://news.com.com/2100-1001-274577.html?legacy=cnet>



Resources

□ News Articles:

- 'Group Pushes Standards For Vulnerability Disclosure', Jaikumar Vijayan, Computerworld, November 19, 2001 - <http://www.computerworld.com/securitytopics/security/story/0,10801,65863,00.html>
- 'No hiding place – The protection of privacy will be a huge problem for the Internet society', The Economist, January 23, 2003 - http://www.economist.com/surveys/displayStory.cfm?story_id=1534283
- 'Powerful attack cripples majority of key Internet computers', Ted Bridis, Associated Press, October 22, 2002 - <http://online.securityfocus.com/news/1400>
- 'Internet's foundations shaken by attack', New Scientist, October 23, 2002 - <http://www.newscientist.com/news/news.jsp?id=ns99992963>
- 'Major Net backbone attack could be first of many', Paul Roberts, Computerworld, October 23, 2002 - <http://www.computerworld.com/securitytopics/security/story/0,10801,75336,00.html>
- 'A net meltdown is inevitable', Bob Alberti, October 25, 2002 - <http://zdnet.com.com/2100-1107-963337.html>
- 'FBI Warns of Chinese Hack Threat', Michelle Delio, Wired News, April 27, 2001 - <http://www.wired.com/news/politics/0,1283,43417,00.html>
- 'US and Chinese hackers trade blows', Mark Ward, BBC News, May 1, 2002 - <http://news.bbc.co.uk/2/hi/science/nature/1306591.stm>
- 'Software Licensing: The Hidden Threat to Information Security', Richard Forno, SecurityFocus, January 23, 2002 - <http://www.securityfocus.com/columnists/55>



Resources

□ News Articles:

- 'Destructive 'ILOVEYOU' computer virus strikes worldwide', D. Ian Hopper, CNN, May 4, 2000 - <http://www.cnn.com/2000/TECH/computing/05/04/iloveyou/>
- 'New 'Love Bug' Computer Virus Sweeps World', Derek Caney, Reuters, May 4, 2000 - <http://www.techtv.com/news/security/story/0,24195,2561838,00.html>
- 'Code Red: Is This The Apocalypse?', Michelle Delio, Wired News, July 30, 2001 - <http://www.wired.com/news/technology/0,1282,45681,00.html>
- 'Net braces for stronger 'Code Red' attack', CNN, July 31, 2001 - <http://www.cnn.com/2001/TECH/internet/07/30/code.red/>
- 'Melissa virus creator jailed', BBC News, May 2, 2002 - <http://news.bbc.co.uk/2/hi/americas/1963371.stm>
- 'Philippine officials charge alleged 'Love Bug' virus creator', CNN, June 29, 2000 - <http://www.cnn.com/2000/TECH/computing/06/29/philippines.lovebug.02/>
- 'Mafiaboy States His Plea', Reuters, August 3, 2000 - <http://www.wired.com/news/politics/0,1283,38011,00.html>
- 'Teen hacker 'Mafiaboy' pleads guilty to 55 charges', Dan Verton, Infoworld, January 18, 2001 - <http://archive.infoworld.com/articles/hn/xml/01/01/18/010118hnmafiaboy.xml>
- 'Mafiaboy Sentenced to 8 Months', Wired News, September 13, 2001 - <http://www.wired.com/news/business/0,1367,46791,00.html>
- 'Feds crack huge identity theft ring', Matt Berger, Infoworld, November 26, 2002 - <http://archive.infoworld.com/articles/hn/xml/02/11/26/021126hnidtheft.xml>



Resources

□ News Articles:

- 'Virus Writer's Conviction Upheld', Joris Evers, PC World, October 28, 2002 - <http://www.pcworld.com/news/article/0,aid,106443,00.asp>
- 'DOJ indicts alleged British hacker', Robert Lemos, C-net News, November 12, 2002 - <http://news.com.com/2100-1001-965490.html>
- 'Massive hacking spree halted; man indicted', Dan Verton, Computerworld, November 12, 2002 - <http://www.computerworld.com/securitytopics/security/hacking/story/0,10801,75833,00.html>
- 'British man indicted for hacking into military networks', Shane Harris, GovExec.com, November 12, 2002 - <http://www.govexec.com/dailyfed/1102/111202h2.htm>
- 'Is Digital Piracy Entering Dark Ages?', Andrew Brandt, PC World, April 19, 2002 - <http://www.pcworld.com/news/article/0,aid,95344,00.asp>
- 'ACLU Acts Against Patriot Act', Julia Scheeres, Wired News, October 16, 2002 - <http://www.wired.com/news/conflict/0,2100,55838,00.html>
- 'ACLU: It's Almost 1984', Wired News, January 15, 2003 - <http://www.wired.com/news/politics/0,1283,57228,00.html>
- 'Man Pleads Guilty to Writing Viruses', Laura Rohde, PC World, December 24, 2002 - <http://www.pcworld.com/news/article/0,aid,108246,00.asp>
- 'Digital Copyright Law Under Scrutiny', Scarlett Pruitt, April 1, 2002 - <http://www.pcworld.com/news/article/0,aid,92164,00.asp>
- 'Update; Thirty Countries Sign Cybercrime Treaty', Rick Perera, Computerworld, November 23, 2001 - <http://www.computerworld.com/securitytopics/security/story/0,10801,66012,00.html>



Resources

□ News Articles:

- 'Defendant Acquitted in DVD Hacking Case', Gillian Law, IDG News, January 7, 2003 - <http://www.pcworld.com/news/article/0,aid,108462,00.asp>
- 'RIAA Silences Security Code Crackers', Sam Costello, PC World, April 26, 2001 - <http://www.pcworld.com/news/article/0,aid,48546,00.asp>
- 'RIAA Sues 532 'John Does'', Paul Roberts, PC World, January 21, 2004 - <http://www.pcworld.com/news/article/0,aid,114387,00.asp>
- 'Hacker's Arrest Prompts Protests Against Adobe', Kim Zetter, PC World, July 20, 2001 - <http://www.pcworld.com/news/article/0,aid,55864,00.asp>
- 'When Love Came to Town: A Virus Investigation', Kim Zetter, PC World, November 13, 2000 - <http://www.pcworld.com/news/article/0,aid,33392,pq,1,00.asp>
- 'Yes, You Are Being Watched', Stephen Lawson, PC World, December 27, 2002 - <http://www.pcworld.com/news/article/0,aid,108121,pq,1,00.asp>
- 'A Truce Over Copy Controls?' Malaika Costello-Dougherty, PC World, January 14, 2003 - <http://www.pcworld.com/news/article/0,aid,108803,00.asp>
- 'Anti-hacking method of full disclosure under attack from a part of the security industry', Stuart McClure & Joel Scambray, Infoworld - <http://archive.infoworld.com/articles/op/xml/00/08/14/000814opswatch.xml>
- 'Can you hack back?', Deborah Radcliff, CNN, June 1, 2000 - <http://www.cnn.com/2000/TECH/computing/06/01/hack.back.idg/index.html>
- 'The Viral Mind: Understanding the Motives of Malicious Coders', D.D. Shelby, SecurityFocus, May 21, 2002 - <http://online.securityfocus.com/infocus/1583>



Resources

□ News Articles:

- 'White Hat v. Black Hat', by the Interpol European Working Group on Information Technology Crime, SC Infosec Opinionwire, December 11, 2002 - http://www.infosecnews.com/opinion/2002/12/11_01.htm
- 'Ohio hacker pleads guilty to data theft', John Nolan, Associated Press, December 19, 2003 - http://www.boston.com/business/technology/articles/2003/12/19/ohio_hacker_pleads_guilty_to_data_theft/
- 'Federal charge filed against Ohio man accused of hacking Acxiom', Caryn Rousseau, Associated Press, August 15, 2003 - <http://www.securityfocus.com/news/6733>
- 'Hacker accessed customer information, Acxiom reports', Associated Press, August 7, 2003 - <http://www.siliconvalley.com/mld/siliconvalley/6484554.htm>
- 'Homeless hacker' says he'll accept plea bargain', Declan McCullagh, C-net News, January 5, 2004 - http://news.com.com/2100-7348_3-5135351.html
- 'Homeless hacker' surrenders', Declan McCullagh, C-net News, September 9, 2003 - http://news.com.com/2100-1009_3-5073426.html
- 'Sentencing date set in nuclear lab hack case', John Leyden, The Register, January 19, 2004 - <http://www.theregister.co.uk/content/55/34972.html>
- 'Teenage hacker faces £21,000 bill', BBC, December 18, 2003 - http://news.bbc.co.uk/2/hi/uk_news/england/london/3331723.stm
- 'Briton pleads guilty to US nuclear lab hacking attack', Bill Goodwin, Computer Weekly, October 31, 2003 - <http://www.computerweekly.com/articles/article.asp?liArticleID=126141&liArticleTypeID=1&liCategoryID=2&liChannelID=28&liFlavourID=1&sSearch=&nPage=1>



Resources

□ News Articles:

- 'Lawmakers aim to get tough on malicious code', William Jackson, Government Computer News, September 10, 2003 - http://www.gcn.com/vol1_no1/daily-updates/23467-1.html
- 'Blaster Investigation Turns Murky', Paul Roberts, PC World, September 5, 2003 - <http://www.pcworld.com/news/article/0,aid,112344,00.asp>
- 'Congress considers cybersecurity legislation', Grant Gross, Infoworld, September 4, 2003 - http://www.infoworld.com/article/03/09/04/HNcybersecuritylaw_1.html
- 'Authorities arrest Minnesota teen in Internet attack', Ted Bridis, Associated Press, August 29, 2003 - <http://www.securityfocus.com/news/6828>
- 'Congress OKs antispam legislation', Declan McCullagh and Paul Festa, C-Net News, December 8, 2003 - http://news.com.com/2100-1028_3-5116940.html
- 'Bush signs bill aimed at controlling spam', Stacey Cowley, Computerworld, December 16, 2003 - <http://computerworld.com/governmenttopics/government/legislation/story/0,10801,88306,00.html>
- 'Spam Is Still Flowing Into E-Mail Boxes', Jonathan Krim, Washington Post, January 6, 2004 - <http://www.washingtonpost.com/ac2/wp-dyn?pagename=article&node=&contentId=A57315-2004Jan5>



Resources

❑ Legislation:

- '1998 Digital Millennium Copyright Act (DMCA) – H.R. 2281', 105th Congress of the United States, January 27, 1998 - <http://thomas.loc.gov/cgi-bin/query/z?c105:H.R.2281.ENR>:
- 'Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism (USA Patriot Act of 2001) – Public Law 107-56', 107th Congress of the United States, October 26, 2001 - http://frwebgate.access.gpo.gov/cgi-bin/getdoc.cgi?dbname=107_cong_public_laws&docid=f:publ056.107.pdf
- 'Cyber Security Enhancement Act of 2002 – H.R. 3482', 107th Congress of the United States - http://frwebgate.access.gpo.gov/cgi-bin/getdoc.cgi?dbname=107_cong_bills&docid=f:h3482eh.txt.pdf (passes as part of the bill creating the Department of Homeland Security)
- 'European Convention on Cybercrime', Council of Europe, November 23, 2001 - <http://conventions.coe.int/Treaty/en/Treaties/Html/185.htm>
- 'Controlling the Assault of Non-Solicited Pornography and Marketing Act of 2003, or the CAN-SPAM Act of 2003' - <http://www.cauce.org/S877.pdf>